Week 1 – Why Data Matters

COMP517: Data Analysis
SCHOOL OF ENGINEERING, COMPUTER AND MATHEMATICAL SCIENCES

This Week

- Introduction to the course
- What is data, information, knowledge?
- Why is Data Analysis important?
- Data Analysis process
- Data Science vs Data Analysis

Course Information

We refer to Canvas pages for:

1. **Course Overview**
   - Course Learning Outcomes
   - Course Content
   - Textbooks and References
2. **Teaching Team**
3. **Assessment Overview**
   - Overall requirement/s to pass the paper:
   - To pass the course, students **must attempt all summative assessments** and achieve a minimum overall grade of C-.
4. **Course Schedule**
5. **Course Descriptor**

Resources

Additional Reference Texts

1. Python Data Analysis : Perform Data Collection, Data Processing, Wrangling, Visualization, and Model Building Using Python , 2021, by Navlani, Avinash, Idris, Ivan, Fandango, Armando

2. The Foundations of Data Science By Ani Adhikari and John DeNero (online Textbook from inferentialthinking.com)

3. Python for Data Analysis : Data Wrangling with Pandas, NumPy, and Ipython , 2018, by McKinney, Wes

Canvas

Check Canvas, twice weekly at least, for:

- Weekly course materials
- Tutorial exercises
- Announcements

Collaboration

Asking questions is greatly encouraged

- ✓ Discuss questions with each other
- ✓ Submit lab tasks individually; discuss in lab
- ✓ Submit Assignment-1 individually, but feel free to discuss
- ✓ Submit Assignment-2 with a partner from your lab;
- ✓ **Note:** only 1 person needs to submit but the second person needs to ensure that it has been submitted.

The Limits of collaboration

- Don't share solutions with each other (except project partners)
- Copying or other dishonesty will result in severe penalties

Getting help

- Post your questions to the discussion forum
- Ask questions during lab sessions
- Ask a classmate

Course Plan

Topics
Introduction to Data Analysis
Data Loading, File Format and Inspection
Exploratory Data Analysis (EDA)
Data Cleaning and preprocessing, Data Aggregation and Correlation
Data Visualization
Time Series analysis
Hypothesis Testing
Statistical foundations: ANOVA
Statistical foundations: Linear Regression
Learning from data

Data

- What is data ?
- "Factual information" (such as measurements or statistics) used as a basis for reasoning, discussion, or calculation"

https://www.merriam-webster.com/dictionary/data

Fact – "data" is plural, "datum" is the singular, so there are ample opportunities to look clever/be annoying about this

The key bit is "factual" – all data use depends on the data being **true and correct**

Data, Information and Knowledge

- Because computing is a relatively young discipline a number of the good words have been used in the language before we got here, so we *overload* words and phrases.
- It’s common to see:
  - Data as 1’s and 0’s in the computer or dimensionless numbers
  - Information – data in context with meaning eg $10 – is that USD, AUD, NZD ?
  - Knowledge – Understanding of meaning - $10 NZD is an expensive cup of coffee.
- The importance of data lies in its ability to provide valuable information to support decision-making processes, by **identifying patterns** and **uncovering trends** to drive innovation.

Why is Data Analysis Important?

Better Customer Targeting:

- You don’t want to waste your business’s precious time, resources, and money putting together **advertising campaigns** targeted at demographic groups that have little to no interest in the goods and services you offer.
- Data analysis helps you see where you should be focusing your advertising efforts.

Why is Data Analysis Important?...

You Will Know Your Target Customers Better:

- Data analysis **tracks** how well your **products and campaigns are performing** within your target demographic.
- Through data analysis, your business can get a better idea of your target audience's **spending habits, disposable income, and most likely areas of interest**

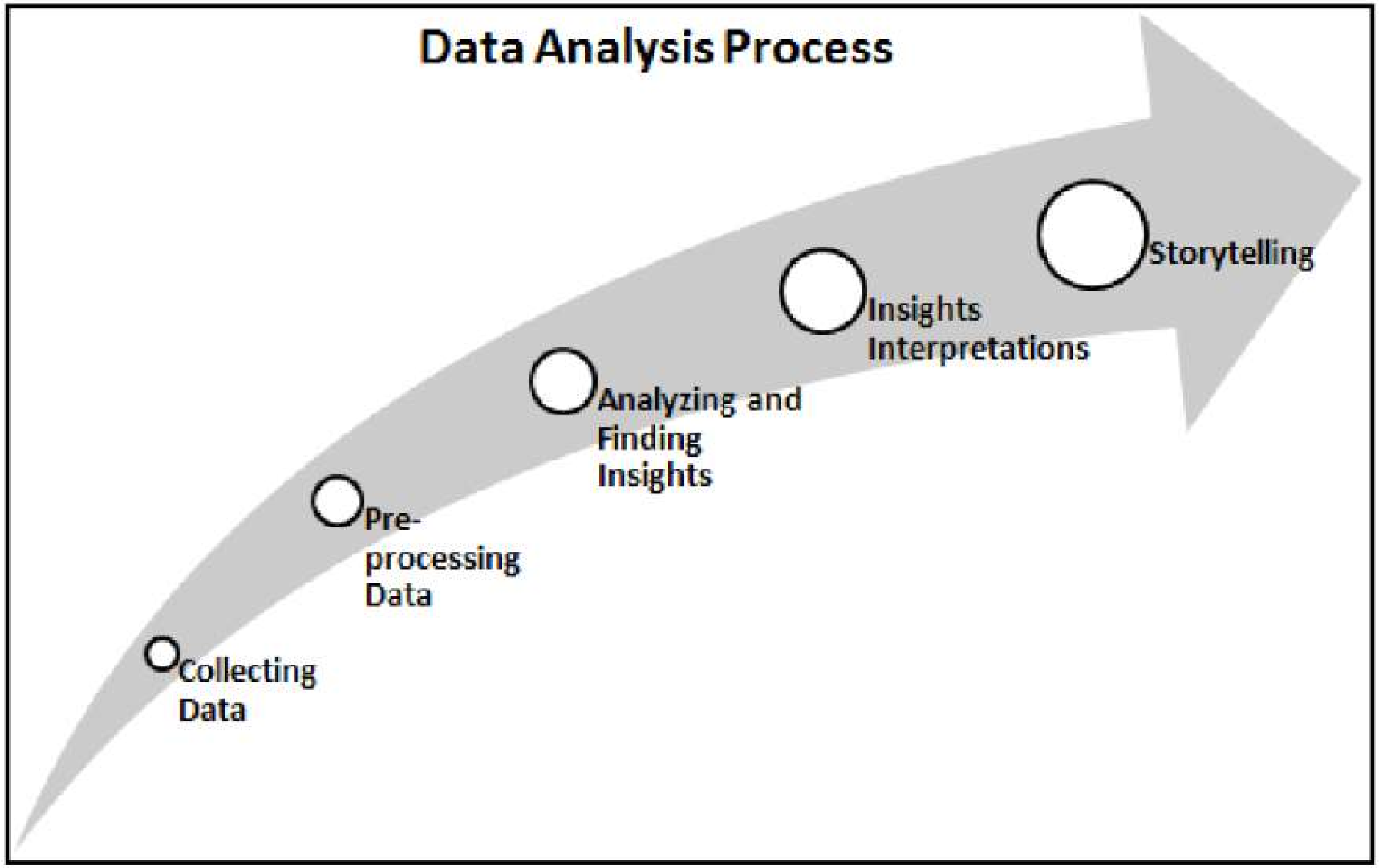
Why is Data Analysis Important?...

Better Problem-Solving Methods:

- Informed decisions are more likely to be successful decisions. Data provides businesses with information. You can see where this progression is leading.
- Data analysis helps businesses make the right choices and avoid costly pitfalls.

Data Analysis Process

Data analysis refers to investigating the data, finding meaningful insights from it, and drawing conclusions.



Data Analysis Process

1. **Collecting Data:** Collect and gather data from several sources.
2. **Preprocessing Data:** Filter, clean, and transform the data into the required format.
3. **Analyzing and Finding Insights:** Explore, describe, and visualize the data and find insights and conclusions.
4. **Insights Interpretations:** Understand the insights and find the impact each variable has on the system.
5. **Storytelling:** Communicate your results in the form of a story so that a layman can understand them.

Knowledge Discovery from Data

- The main objective of KDD is to extract or discover hidden interesting patterns from large databases, data warehouses, and other web and information repositories.
    1. Data Cleaning
    2. Data Integration
    3. Data Selection
    4. Data Transformation
    5. Data Mining
    6. Pattern Evaluation
    7. Knowledge Presentation.

Data Analysis and Data Science

- **Data Science** is an interdisciplinary area that uses a scientific approach to extract insights from structured and unstructured data.
  - It is a union of all terms, including data analytics, data mining, machine learning, and other related domains.
  - It is not only limited to exploratory data analysis and is used for developing models and prediction algorithms such as stock price, weather, disease, fraud forecasts, and recommendations such as movie, book, and music recommendations.

- **Data Analysis** is the process in which data is explored in order to discover patterns that help us make business decisions.
  - subdomains of data science
  - performs exploratory data analysis using descriptive statistics, visualizes data, performs statistical analysis, prepares reports, and presents it to the business decision making authorities.

Roles of Data Scientist & Data Analyst

- Data scientists focus on **what is going to happen** , whereas data analysts focus on **what has happened so far** .

Features	Data Analyst	Data Scientist
Background	Discover meaningful insights from the data	Predict future events and scenarios based on data
Role	Solve the business questions to make decisions.	Formulate questions that can profit the business
Skillset	Knowledge of statistics, SQL, and data visualization	Knowledge of statistics, machine learning algorithms, NLP, and deep learning

Chapter 1 of Navlani textbook
https://ebookcentral.proquest.com/lib/aut/reader.action?docID=6462897

Types of Data Analysis

Diagnostic Analysis:

- Diagnostic analysis answers the question, "**Why did this happen?**"
- Using insights gained from statistical analysis (more on that later!), analysts use diagnostic analysis to **identify patterns in data**.
- Ideally, the analysts find similar patterns that existed in the past, and consequently, use those to resolve the present challenges, hopefully.

Types of Data Analysis…

Predictive Analysis:

- Predictive analysis answers the question, “**What is most likely to happen?**”
  - By using patterns found in older data as well as current events, analysts predict future events.
- While there’s no such thing as 100 percent accurate forecasting, the odds improve if the analysts:
  - have plenty of detailed information and
  - the discipline to research it thoroughly.

Types of Data Analysis…

Prescriptive Analysis:

- An advanced form of data analytics that
  - not only predicts future outcomes but also
  - **recommends actions to achieve desired results** or prevent undesirable ones.
- Mix all the insights gained from the other data analysis types, and you have prescriptive analysis.
- Sometimes, an issue can’t be resolved solely with one analysis type, but instead, requires multiple insights.
- Uses various statistical techniques from predictive modelling, as well as artificial intelligence (AI) and machine learning (ML) algorithms.

Types of Data Analysis…

Quantitative Data Analysis: Statistical data analysis methods collect raw data and process it into numerical data.

Quantitative analysis methods include:

- **Hypothesis Testing,** for assessing the truth of a given hypothesis or theory for a data set or demographic.
- **Mean, or average** determines a subject's overall trend by dividing the sum of a list of numbers by the number of items on the list.

The Key is USING the data

- In general, **if** it cannot influence your decision (or answer your question) then using data is probably not worth it.
- This means the question should to be sensible or at least answerable in some way using data.

- Examples – most popular biscuit in New Zealand
- “Which is the best biscuit?”. Your answer?
- Will the outcome change whether you like a particular biscuit or not?

On biscuits..

https://www.thehits.co.nz/the-latest/new-zealands-biscuit-of-the-year-has-been-revealed-and-its-very-controversial/

Computers and data

- Generally computers store data as “binary” states either a 1 or a 0.
- This can be a “high” or “low” voltage (normally about 8v is high in PC’s
- It can also be off or on for a light source, or different frequencies or amplitudes for radio waves, static charge, or magnetic polarity
- https://www.youtube.com/watch?v=wteUW2sL7bc&ab_channel=TED-Ed

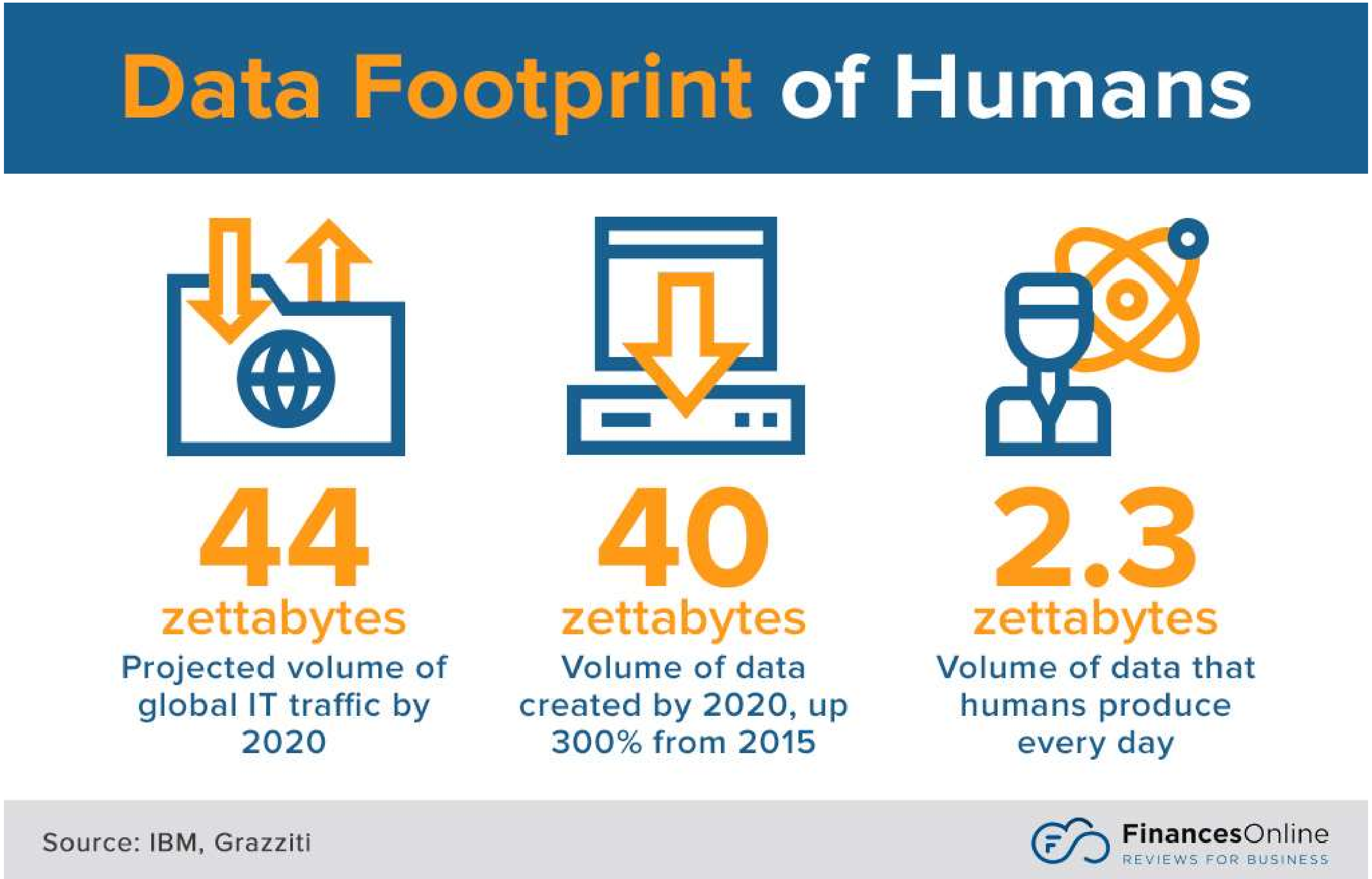
Some numbers

- A bit is a 1 or a 0
- A byte is 8 bits
- And then.. (thanks to Wiki)

Value	Metric	
1000	kB	kilobyte
1000^2	MB	**megabyte**
1000^3	GB	gigabyte
1000^4	TB	terabyte
1000^5	PB	petabyte
1000^6	EB	exabyte
1000^7	ZB	zettabyte
1000^8	YB	yottabyte

Data now

- The 2.5 quintillion bytes of data we create every day is useless unless we crunch it using big data analytics.



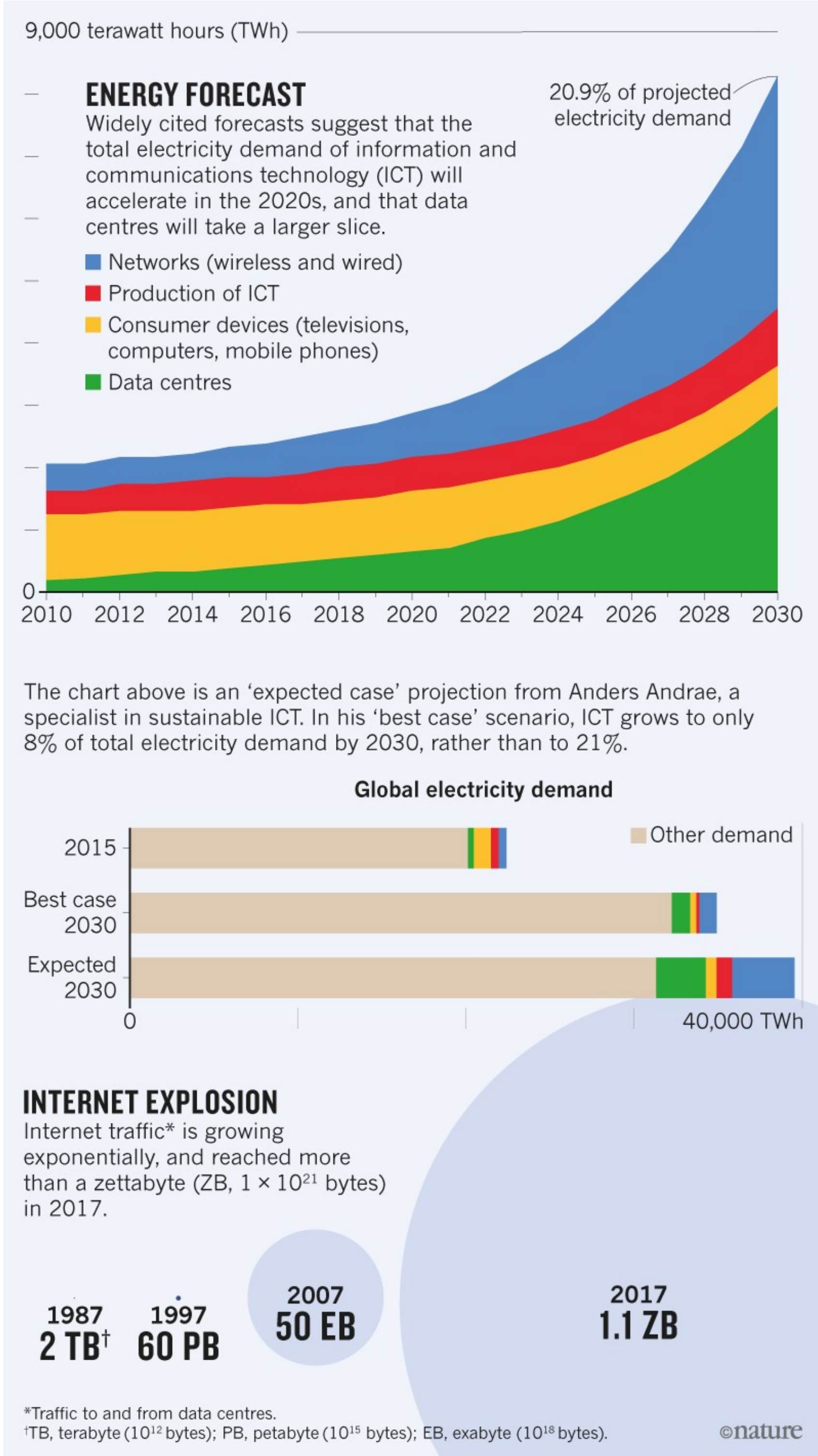
https://financesonline.com/big-data-statistics/

The cost

- Carbon emissions:

https://www.nature.com/articles/d41586-018-06610-y

- In some cases 80% of the data is duplicated



Some data flows

- Netflix – about 1 GB per hour for standard movies
- Home use https://www.chorus.co.nz/data-calculator#/
- Facebook – around 2 Billion users, around 4 petabyte generated a day
- SKA 8 TB/second is processed – around a half a million times /day (it is not all stored)
- SKA Observatory https://www.skatelescope.org/software-and-computing/

Data Journalism

- Increasingly people are using data to uncover/report news stories; using data to "inform" advertisements, etc.
- What *exactly* they mean is up for debate sometimes
- Example of a Data Journalism website:

  https://www.dw.com/en/data/t-43091100

- The Global Investigative Journalism Network (GIJN)
  - How AI is impacting Press Freedom and Investigative Journalism

https://gijn.org/stories/ai-impacts-press-freedom-investigative-journalism/

COVID Vaccines Effectiveness

- Say 100,000 people have had the vaccine and 100,000 have not.

Size of outbreak	Number of unvaccinated people who will get COVID	Number of Vaccinated people who will get COVID
Small	20	1
Medium	1000	50
Huge (complete transmission)	100,000	5,000

More numbers

- There is some evidence that vaccines also reduce the chance of transmitting COVID
- https://www.bmj.com/content/373/bmj.n888
- Mortality figures for COVID https://www.worldometers.info/coronavirus/
- Good article by David Spiegelhalter at:
- https://www.theguardian.com/profile/david-spiegelhalter

Some interesting websites

- Gapminder https://www.gapminder.org/tools/#$chart-type=bubbles&url=v1
- https://www.worldometers.info/coronavirus/

Take home messages

- Be on the lookout for data and how we can use it
- Data needs to be managed and understood before it can be analysed